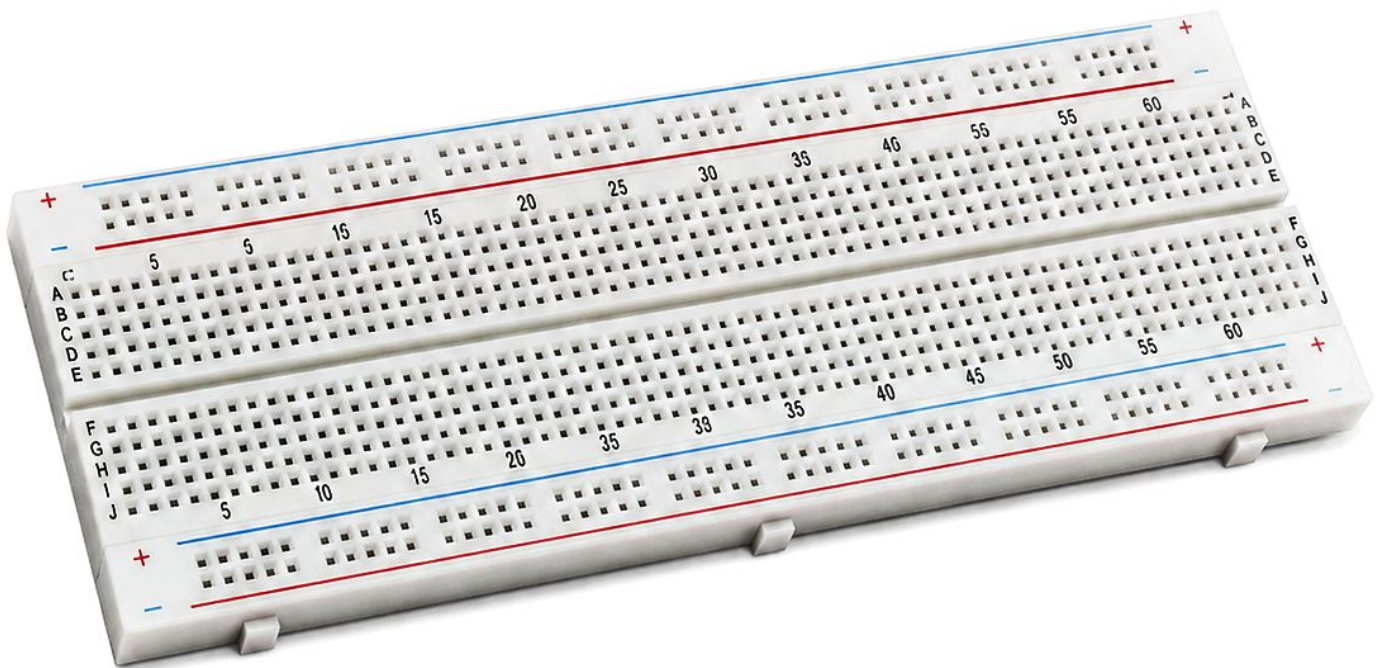


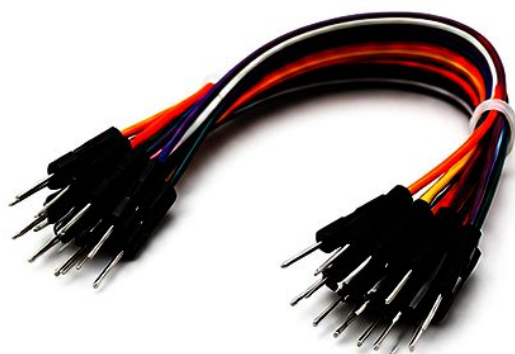


BREAD BOARD



JUMPER WIRES

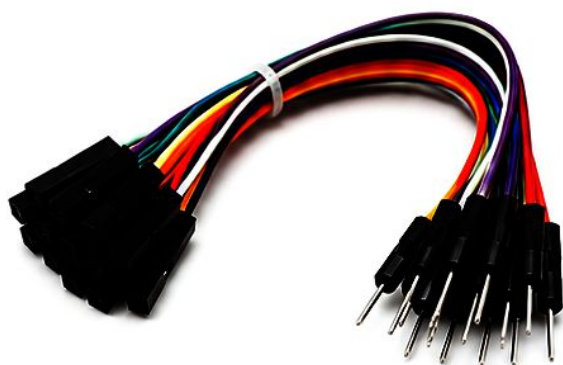
MALE TO MALE WIRE



MALE TO FEMALE WIRE



FEMALE TO MALE WIRE



FEMALE TO FEMALE WIRE



LEDs

BRIGHT • EFFICIENT • RELIABLE

COLOURFUL LEDs

RED



BLUE



GREEN



WHITE



MINI BULB LEDs



RED MINI BULB



BLUE MINI BULB



GREEN MINI BULB



WHITE MINI BULB



LOW POWER
CONSUMPTION



HIGH
BRIGHTNESS



LONG LIFE &
DURABLE



ECO
FRIENDLY

RESISTORS

HIGH QUALITY • PRECISE VALUES • RELIABLE PERFORMANCE

DIFFERENT OHM VALUES

100 Ω



Brown • Black • Brown • Gold
($\pm 5\%$)

220 Ω



Red • Red • Brown • Gold
($\pm 5\%$)

330 Ω



Orange • Orange • Brown • Gold
($\pm 5\%$)

470 Ω



Yellow • Violet • Brown • Gold
($\pm 5\%$)

1 K Ω



Brown • Black • Red • Gold
($\pm 5\%$)

2.2 K Ω



Red • Red • Red • Gold
($\pm 5\%$)

4.7 K Ω



Yellow • Violet • Red • Gold
($\pm 5\%$)

10 K Ω



Brown • Black • Orange • Gold
($\pm 5\%$)

22 K Ω



Red • Red • Orange • Gold
($\pm 5\%$)

47 K Ω



Yellow • Violet • Orange • Gold
($\pm 5\%$)

100 K Ω



Brown • Black • Yellow • Gold
($\pm 5\%$)

220 K Ω



Red • Red • Yellow • Gold
($\pm 5\%$)

470 K Ω



Yellow • Violet • Yellow • Gold
($\pm 5\%$)

1 M Ω



Brown • Black • Green • Gold
($\pm 5\%$)

10 M Ω



Brown • Black • Blue • Gold
($\pm 5\%$)



**PRECISE
TOLERANCE**
 $\pm 5\%$



**HIGH
STABILITY**



**WIDE RANGE
OF VALUES**



**PREMIUM
QUALITY**



**ROHS
COMPLIANT**

CAPACITORS

HIGH QUALITY • STABLE PERFORMANCE • LONG LIFE

DIFFERENT VALUES (FARADS)

CERAMIC CAPACITORS

10 pF  $10 \times 10^{-12} \text{ F}$ (10pF)	22 pF  $22 \times 10^{-12} \text{ F}$ (22pF)	47 pF  $47 \times 10^{-12} \text{ F}$ (47pF)	100 pF  $100 \times 10^{-12} \text{ F}$ (100pF)	220 pF  $220 \times 10^{-12} \text{ F}$ (220pF)	470 pF  $470 \times 10^{-12} \text{ F}$ (470pF)	1 nF  $1 \times 10^{-9} \text{ F}$ (1nF)	4.7 nF  $4.7 \times 10^{-9} \text{ F}$ (4.7nF)	10 nF  $10 \times 10^{-9} \text{ F}$ (10nF)	100 nF  $100 \times 10^{-9} \text{ F}$ (100nF)
--	---	---	--	--	--	---	---	--	---

ELECTROLYTIC CAPACITORS

1 μF  $1 \times 10^{-6} \text{ F}$ (1 μF)	2.2 μF  $2.2 \times 10^{-6} \text{ F}$ (2.2 μF)	4.7 μF  $4.7 \times 10^{-6} \text{ F}$ (4.7 μF)	10 μF  $10 \times 10^{-6} \text{ F}$ (10 μF)	22 μF  $22 \times 10^{-6} \text{ F}$ (22 μF)	47 μF  $47 \times 10^{-6} \text{ F}$ (47 μF)	100 μF  $100 \times 10^{-6} \text{ F}$ (100 μF)	220 μF  $220 \times 10^{-6} \text{ F}$ (220 μF)	470 μF  $470 \times 10^{-6} \text{ F}$ (470 μF)	1000 μF  $1000 \times 10^{-6} \text{ F}$ (1000 μF)
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FILM CAPACITORS

10 nF  $10 \times 10^{-9} \text{ F}$ (10nF)	47 nF  $47 \times 10^{-9} \text{ F}$ (47nF)	100 nF  $100 \times 10^{-9} \text{ F}$ (100nF)	220 nF  $220 \times 10^{-9} \text{ F}$ (220nF)	470 nF  $470 \times 10^{-9} \text{ F}$ (470nF)	1 μF  $1 \times 10^{-6} \text{ F}$ (1 μF)	2.2 μF  $2.2 \times 10^{-6} \text{ F}$ (2.2 μF)	4.7 μF  $4.7 \times 10^{-6} \text{ F}$ (4.7 μF)	10 μF  $10 \times 10^{-6} \text{ F}$ (10 μF)
---	--	---	---	---	--	--	--	---



HIGH QUALITY
COMPONENTS



STABLE & RELIABLE
PERFORMANCE



WIDE RANGE
OF VALUES



LONG LIFE &
DURABILITY



LOW LOSS &
HIGH EFFICIENCY



ROHS
COMPLIANT

BUZZER



DIODES

HIGH QUALITY • RELIABLE • EFFICIENT

TYPES OF DIODES

1N4007
RECTIFIER DIODE



V_{RRM} : 1000V
 $I_{F(AV)}$: 1A

1N4148
SIGNAL DIODE



V_{RRM} : 100V
 $I_{F(AV)}$: 150mA

1N5819
SCHOTTKY DIODE



V_{RRM} : 40V
 $I_{F(AV)}$: 1A

1N5408
HIGH CURRENT DIODE



V_{RRM} : 1000V
 $I_{F(AV)}$: 3A

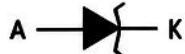
FR107
FAST RECOVERY DIODE



V_{RRM} : 1000V
 $I_{F(AV)}$: 1A
 t_{rr} : 500ns

SPECIALTY DIODES

ZENER DIODE
(BZX55C5V1)



V_Z : 5.1V
 P_D : 500mW

LED (5mm)
LIGHT EMITTING DIODE



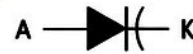
V_F : 1.8 – 2.2V
 I_F : 20mA

PHOTODIODE
(BPW34)



V_R : 60V
 I_{PH} : 50 μ A (typ.)

VARACTOR DIODE
(BBY52)



V_R : 30V
 C_j : 2 – 22pF

TVS DIODE
(SMBJ5.0A)



V_{BR} : 5.0V
 P_{PP} : 600W

APPLICATIONS

- Rectification
- Voltage Regulation
- Switching
- Protection Circuits
- Signal Processing
- Indicators & Lighting



HIGH
EFFICIENCY



RELIABLE
PERFORMANCE



WIDE OPERATING
RANGE



DURABLE &
LONG LIFE



RoHS
COMPLIANT

PUSH BUTTONS & ON / OFF BUTTONS

RELIABLE • DURABLE • EASY TO USE

PUSH BUTTONS

Tactile Push Button (6x6mm)



- Type : Momentary
- Rating : 12V DC, 50mA
- Size : 6 x 6 x 4.3 mm

Tactile Push Button (12x12mm)



- Type : Momentary
- Rating : 12V DC, 50mA
- Size : 12 x 12 x 7.3 mm

Round Push Button (Red)



- Type : Momentary
- Rating : 12V DC, 1A
- Mounting Hole : 12mm

Round Push Button (Green)



- Type : Momentary
- Rating : 12V DC, 1A
- Mounting Hole : 12mm

Large Push Button (Red)



- Type : Momentary
- Rating : 250V DC, 3A
- Mounting Hole : 16mm

Metal Push Button (Momentary)



- Type : Momentary
- Rating : 250V AC, 3A
- Mounting Hole : 16mm

ON / OFF BUTTONS

Rocker Switch (ON / OFF)



- Rating : 6A / 250V AC
- Type : 2 Pin
- Function : ON / OFF

Rocker Switch (ON / OFF) Red



- Rating : 6A / 250V AC
- Type : 2 Pin
- Function : ON / OFF

Mini Toggle Switch (ON / OFF)



- Rating : 6A / 125V AC
- Type : 2 Pin
- Function : ON / OFF

Toggle Switch (ON / OFF)



- Rating : 10A / 250V AC
- Type : 2 Pin
- Function : ON / OFF

Slide Switch (ON / OFF)



- Rating : 0.5A / 50V DC
- Type : 2 Pin
- Function : ON / OFF

Key Switch (ON / OFF)



- Rating : 5A / 250V AC
- Type : 2 Pin
- Function : ON / OFF



EASY TO OPERATE



DURABLE & LONG LASTING



HIGH PERFORMANCE



STABLE & RELIABLE

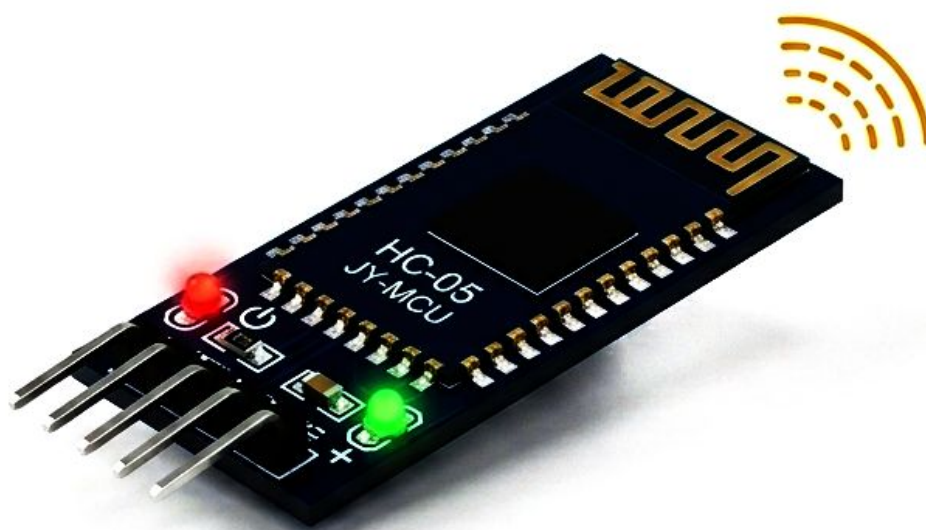


WIDE RANGE OF OPTIONS



SAFE & ECO FRIENDLY

BLUETOOTH MODULE HC05



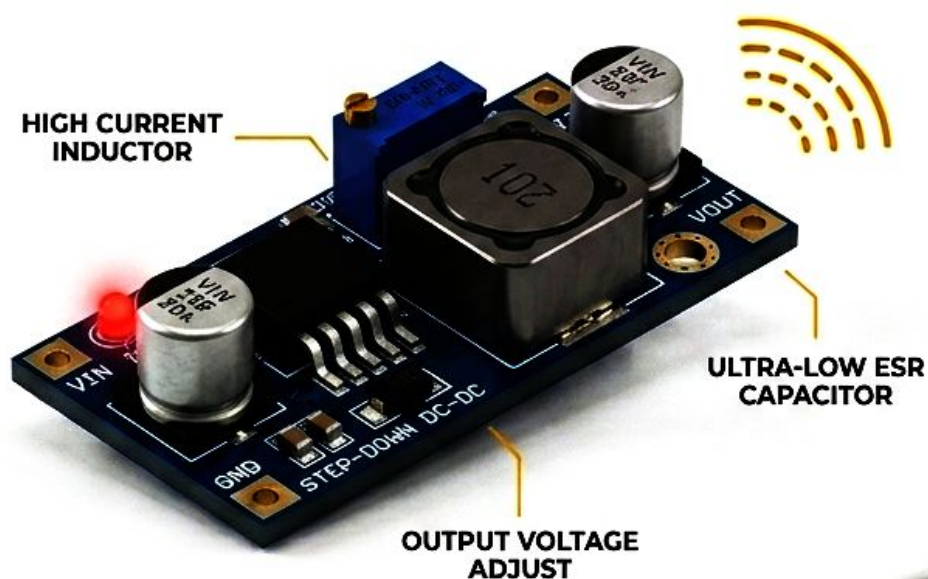


GSM MODULE 800L

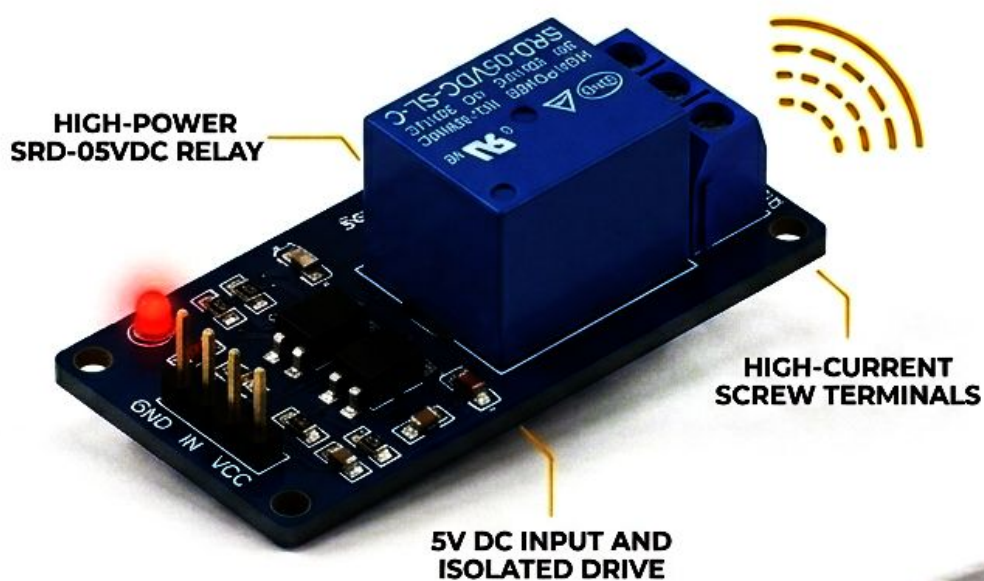




BUCK CONVERTER MODULE



5V RELAY MODULE



BOOST CONVERTER



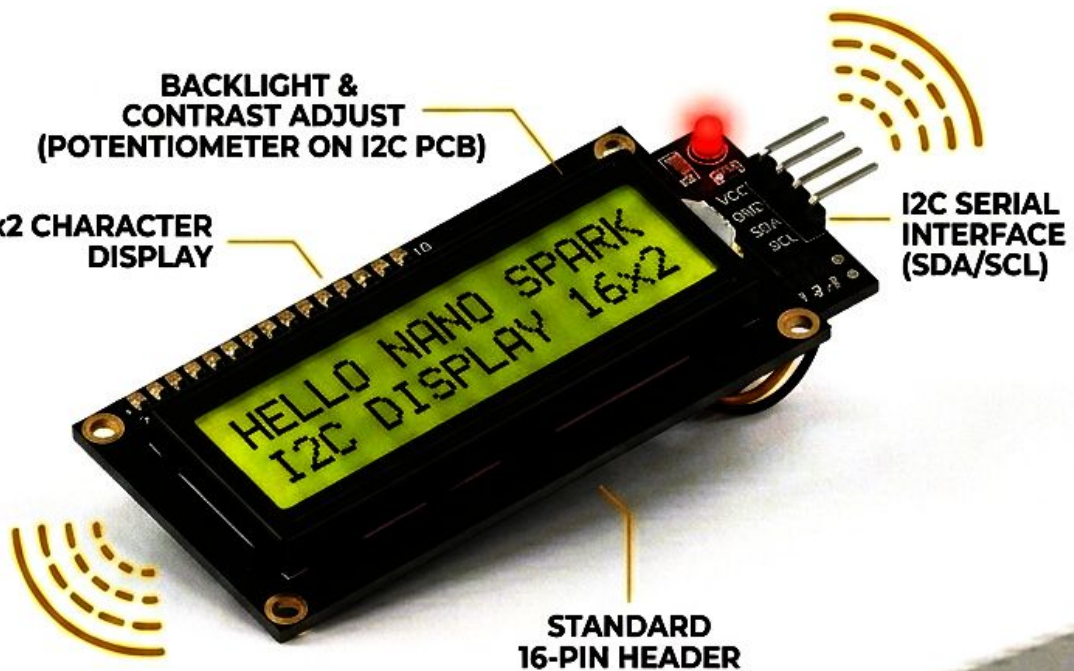
LCD DISPLAY

BACKLIGHT &
CONTRAST ADJUST
(POTENTIOMETER ON I2C PCB)

16x2 CHARACTER
DISPLAY

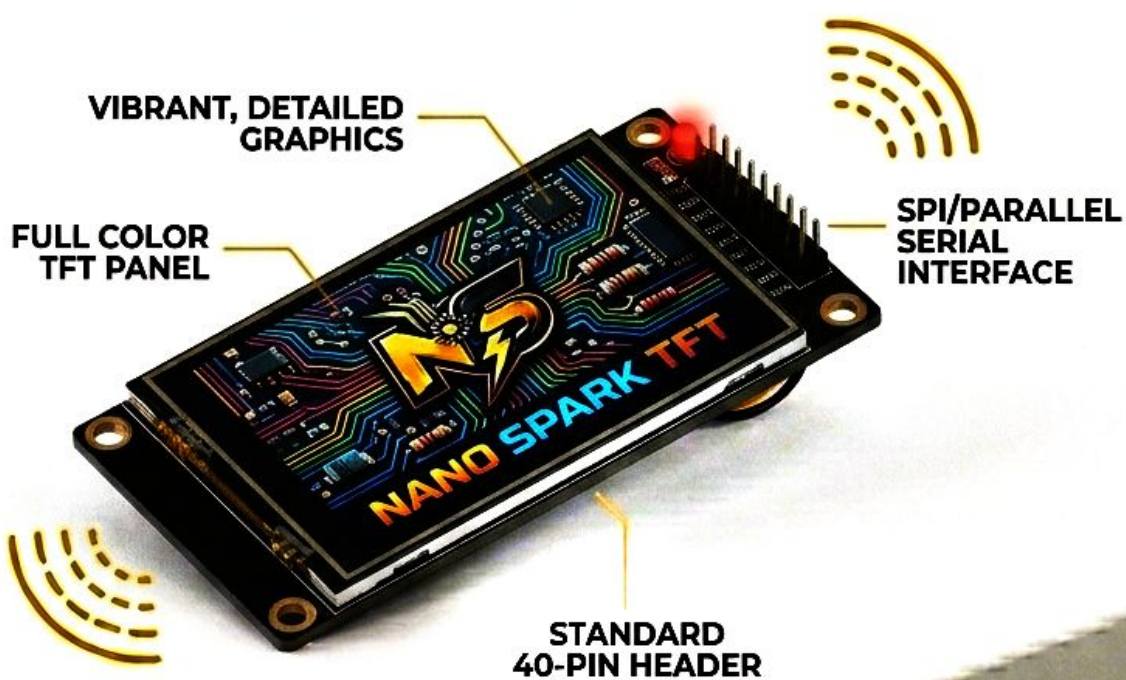
I2C SERIAL
INTERFACE
(SDA/SCL)

STANDARD
16-PIN HEADER



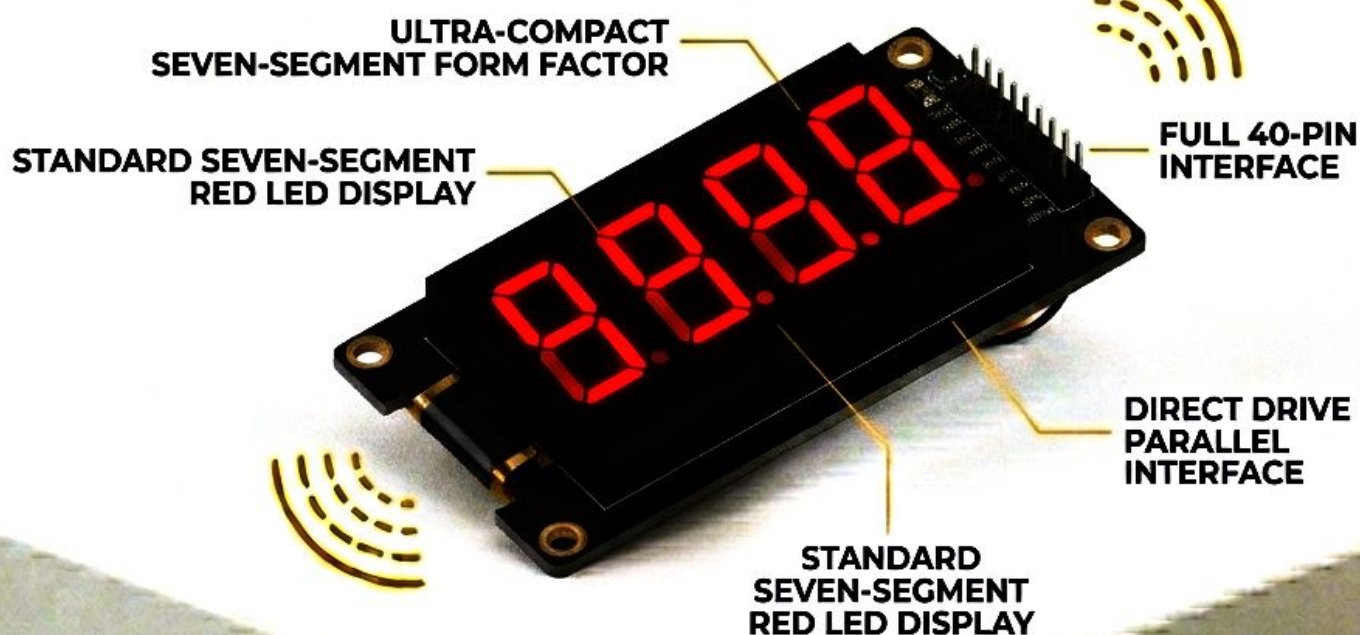


TFT DISPLAY



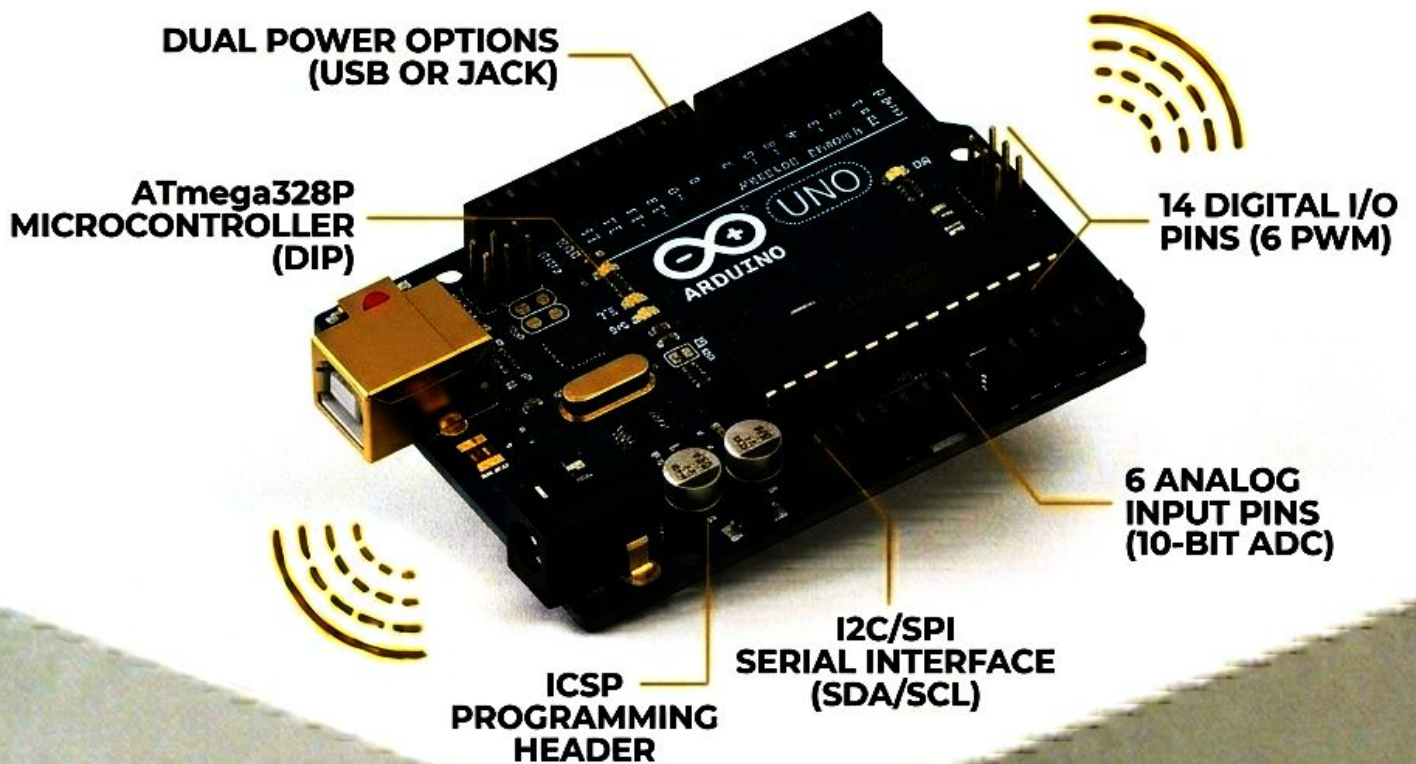


ULTRA-BRIGHT SEVEN-SEGMENT DISPLAY (40-PIN)





ARDUINO UNO R3 MICROCONTROLLER





NANO SPARK
TECHNOLOGY · INNOVATIONS · SOLUTIONS

0.96" OLED DISPLAY MODULE

I2C INTERFACE | 128x64 PIXELS



HIGH
CONTRAST



WIDE VIEWING
ANGLE



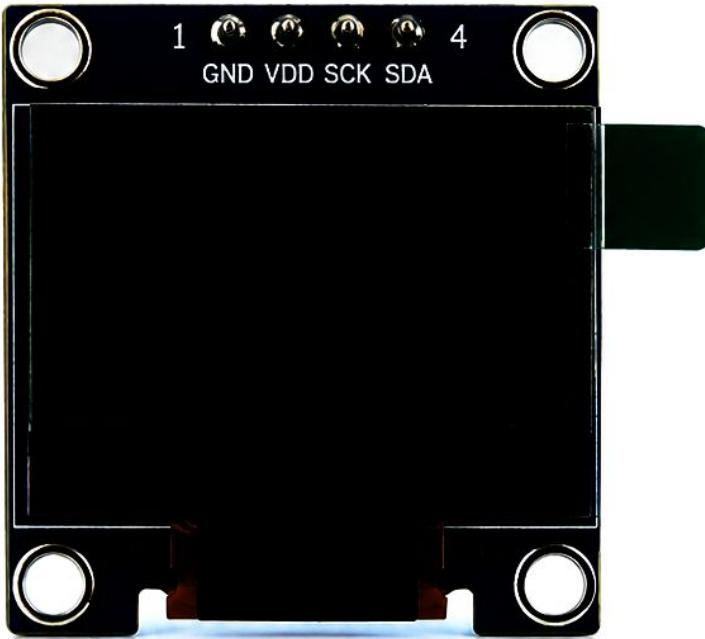
LOW POWER
CONSUMPTION



COMPACT
SIZE



PLUG &
PLAY



SPECIFICATIONS

Display Type	OLED (Organic LED)
Display Size	0.96 inch
Resolution	128 x 64 Pixels
Color	White
Interface	I2C (IIC)
I2C Address	0x3C (Default)
Operating Voltage	3.3V ~ 5V DC
Logic Voltage	3.3V / 5V Compatible
Current Consumption	~20mA (Typical)
Viewing Angle	> 160°
Display Area	21.74mm x 10.86mm
Pixel Size	0.153 x 0.153 mm
Module Size	27mm x 27mm
Operating Temperature	-40°C to +80°C
Storage Temperature	-40°C to +85°C

PIN DESCRIPTION

PIN	NAME	DESCRIPTION
1	GND	Ground (0V)
2	VDD	Power Supply (3.3V – 5V DC)
3	SCK	I2C Serial Clock
4	SDA	I2C Serial Data

INTERFACE (I2C)



DIMENSIONS



APPLICATIONS



IOT DEVICES



WEARABLE DEVICES



HOME AUTOMATION



MEASUREMENT DEVICES



DIY PROJECTS

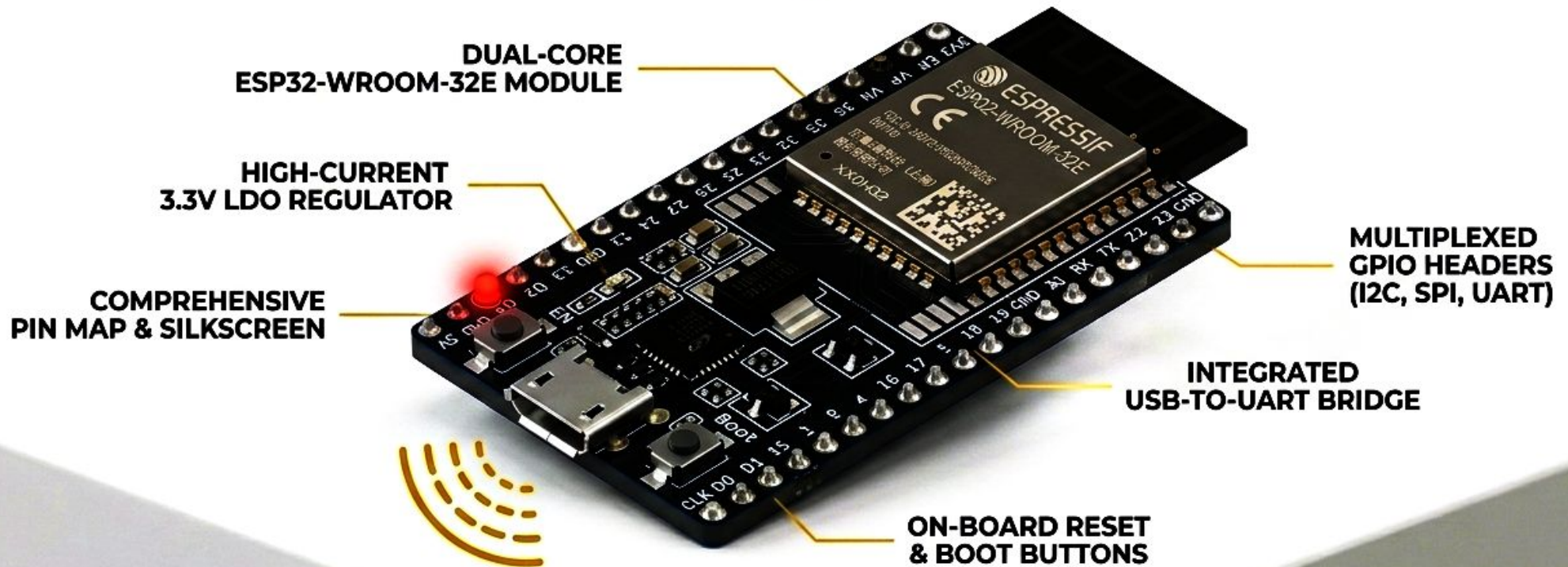


ROBOTICS



PORTABLE DEVICES

ESP32 IoT DEVELOPMENT BOARD



ESP8266 WiFi DEVELOPMENT BOARD



**ULTRA-LOW POWER
ESP8266 SoC & WiFi MODULE**

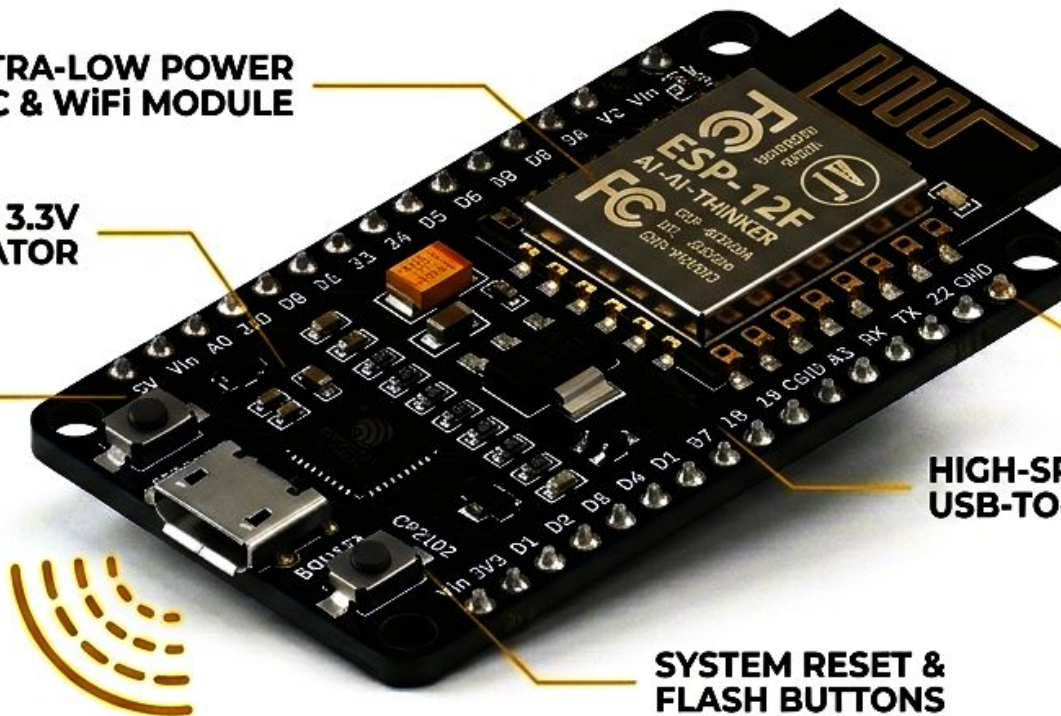
**STABLE 3.3V
LDO REGULATOR**

**COMPLETE PINOUT
& SILKSCREEN MAP**

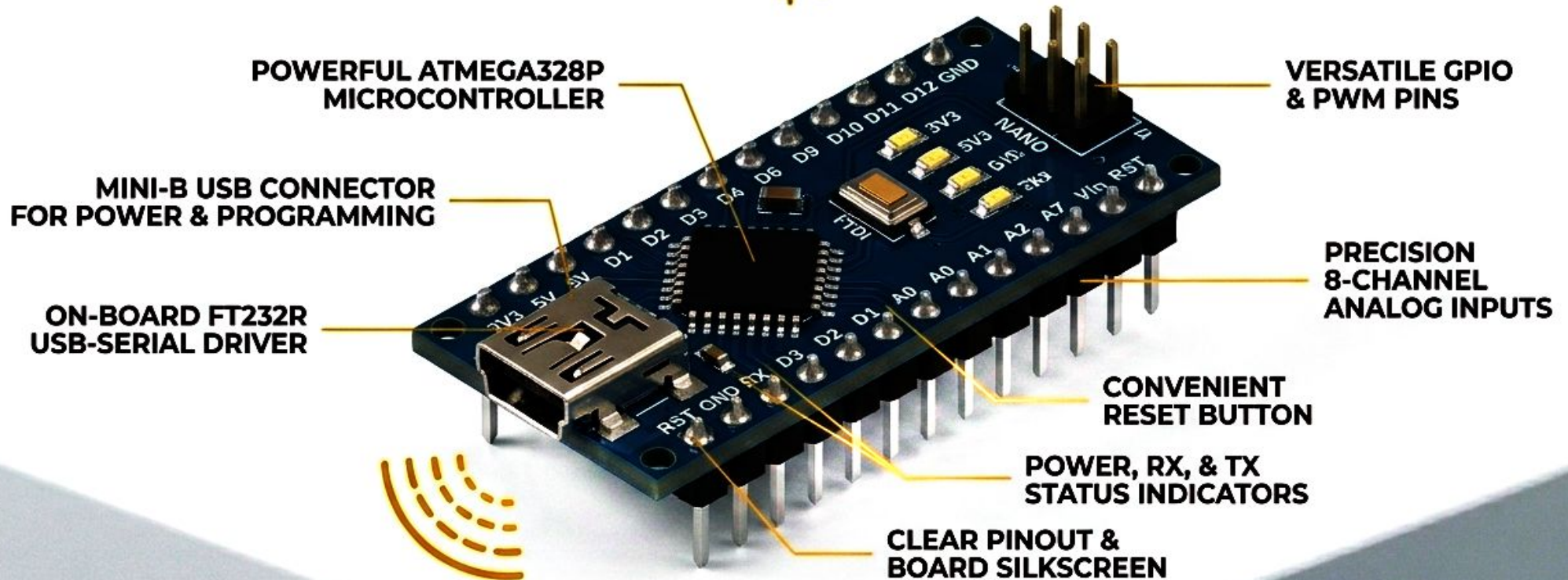
**RECONFIGURABLE
GPIO HEADERS
(UART, I2C, PWM)**

**HIGH-SPEED
USB-TO-UART INTERFACE**

**SYSTEM RESET &
FLASH BUTTONS**



ARDUINO NANO BOARD



DHT11 SENSOR



TEMPERATURE
OUTPUT

HUMIDITY
OUTPUT

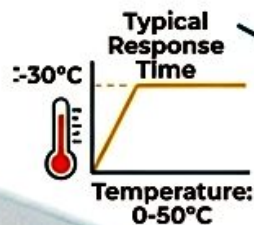
SINGLE-BUS
DIGITAL SIGNAL

LOW-COST
SOLUTION

PRE-CALIBRATED

PRE--CALIBRATED

OPERATING VOLTAGE
VOLTAGE: 3.3V-5.5V



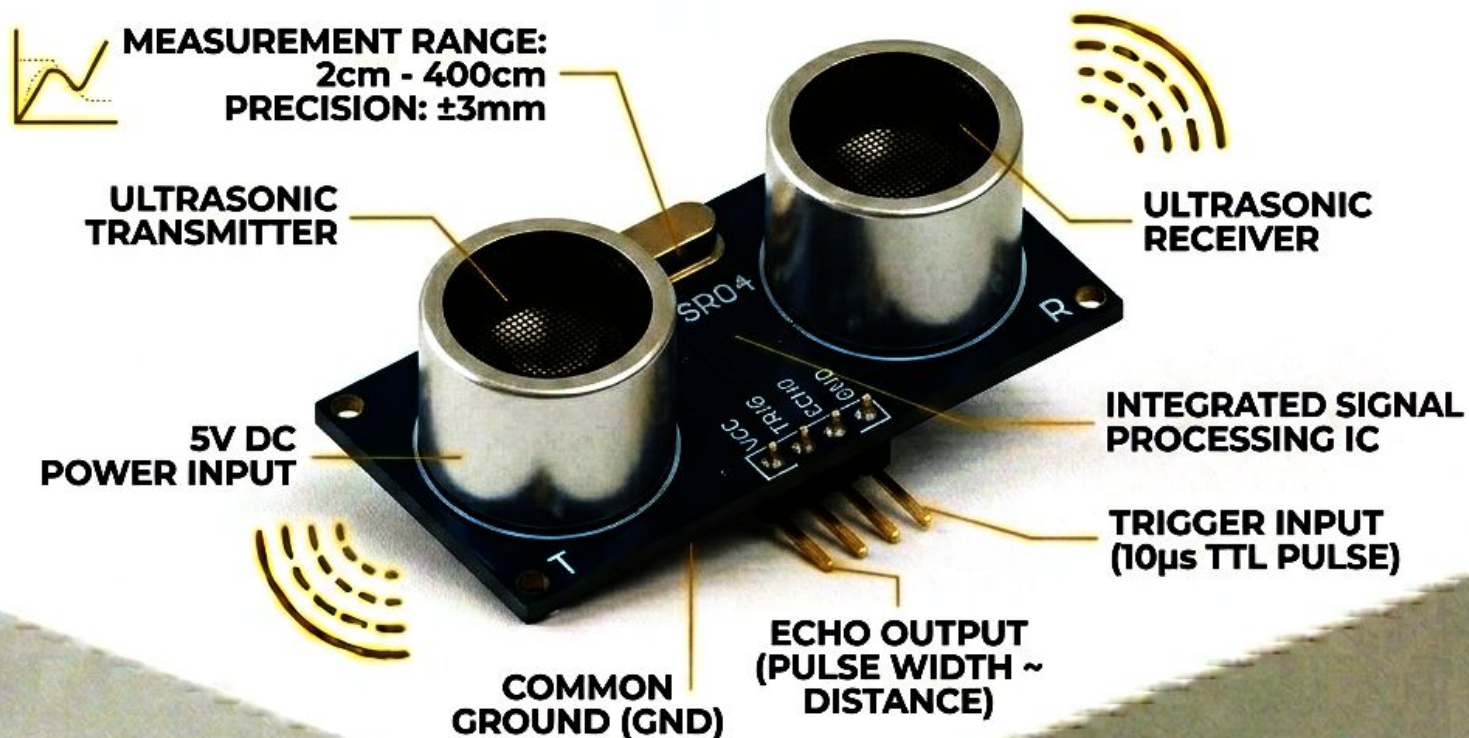
Humidity

Temperature Range:
0-50°C $\pm 2^{\circ}\text{C}$
Humidity: Accurate:
20-90% RH $\pm 5\%$ RH





ULTRASONIC DISTANCE SENSOR HC-SR04





INFRARED (IR) PROXIMITY SENSOR MODULE

SPECIFICATIONS:

- DETECTION RANGE: 2cm - 30cm
- ADJUSTABLE SENSITIVITY
- OPERATING VOLTAGE: 3.3V-5V DC

IR TRANSMITTER
(EMITTER LED)

IR RECEIVER
(PHOTOTRANSISTOR)

POWER INPUT
(VCC)

ON-BOARD SIGNAL
PROCESSING IC

DIGITAL OUTPUT
(OUT)

COMMON
GROUND (GND)

ADJUSTABLE
SENSITIVITY
POTENTIOMETER





FIRE SENSOR MODULE



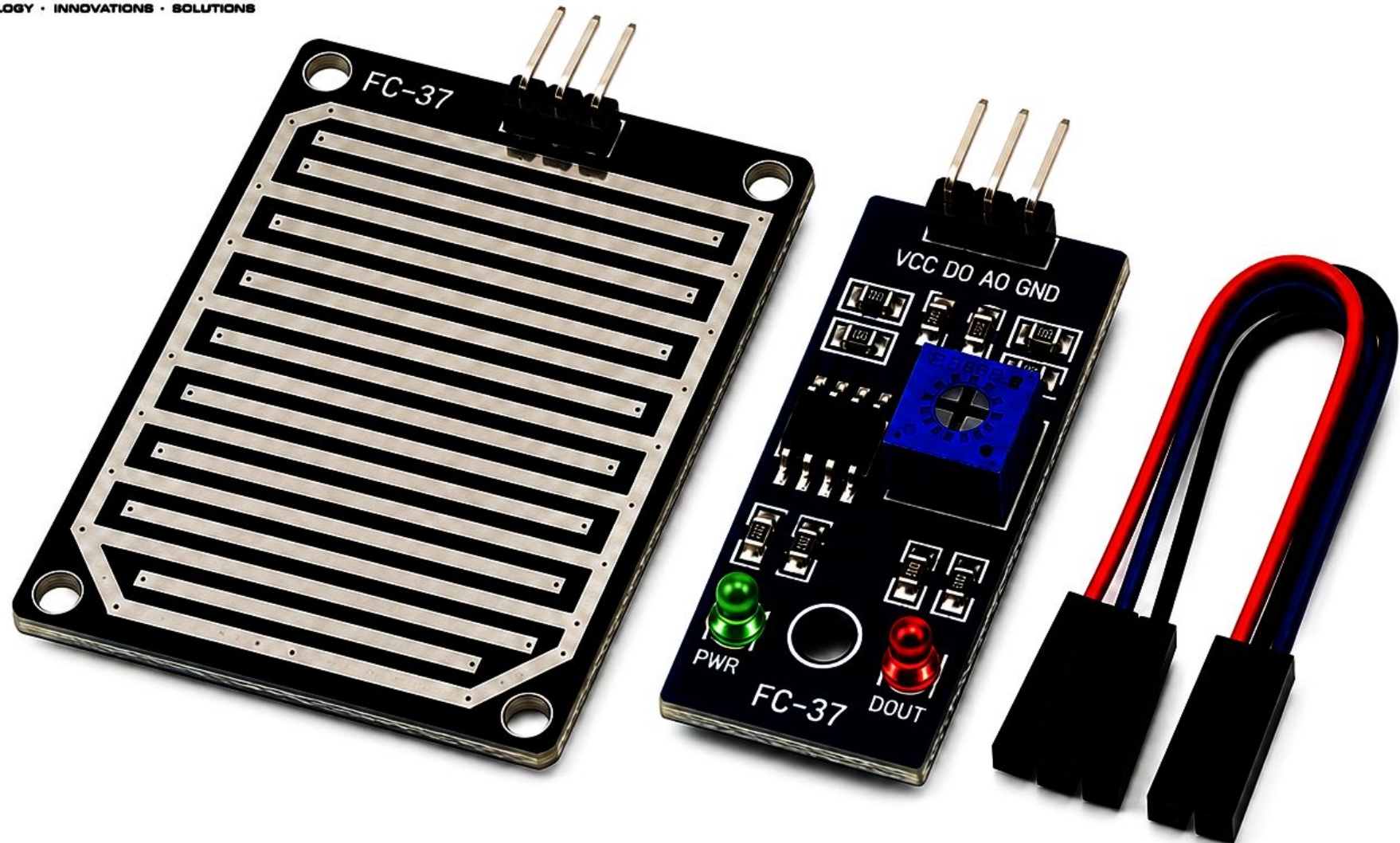
MQ-3 GAS SENSOR MODULE



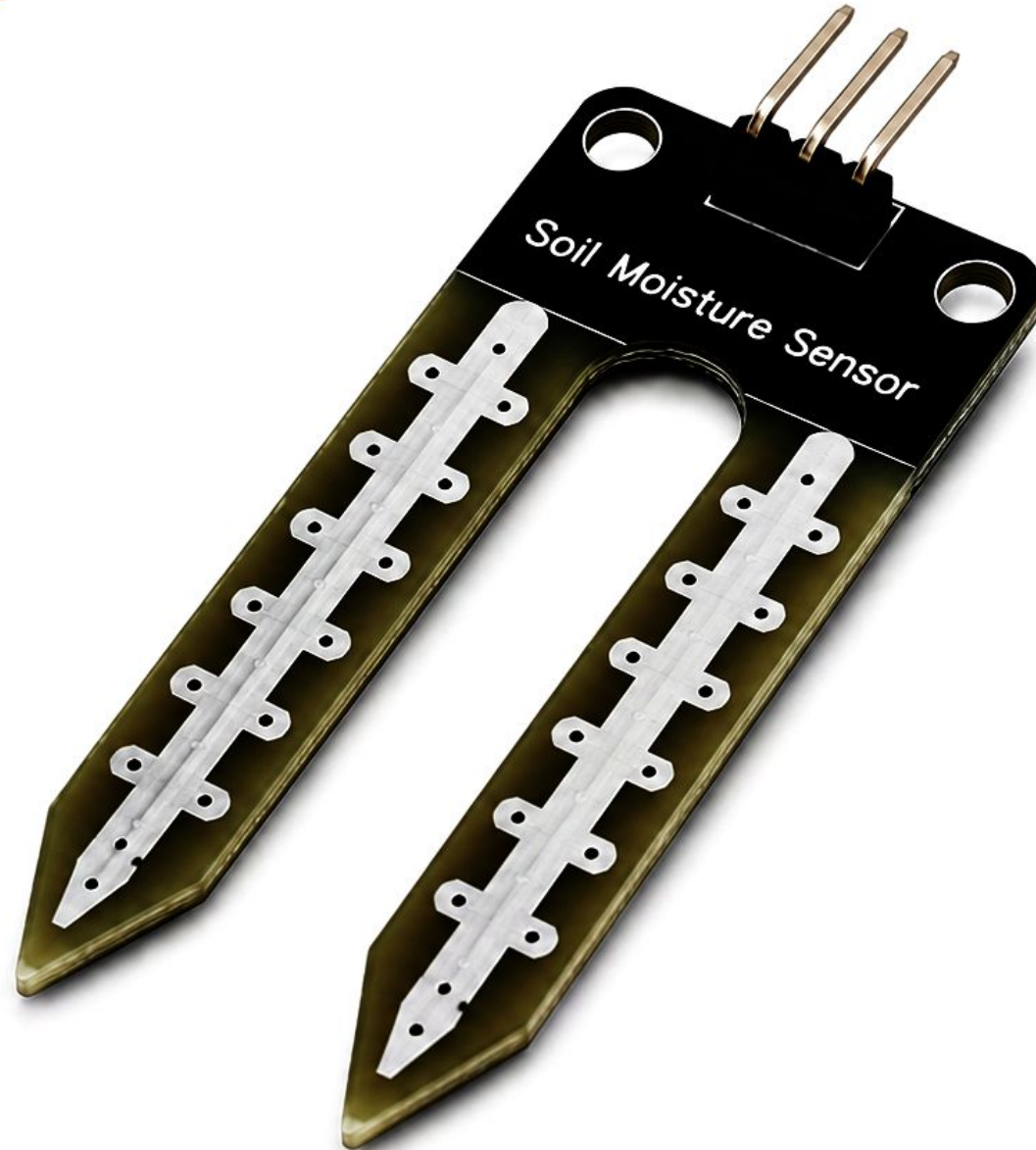
MQ-2 GAS SENSOR MODULE



RAIN SENSOR MODULE



SOIL MOISTURE SENSOR



RFID CARD

13.56MHz | MIFARE CLASSIC 1K | REWRITABLE | CONTACTLESS



**CONTACTLESS
TECHNOLOGY**



**SECURE &
RELIABLE**



**REWRITABLE
& REUSABLE**



**PRE-EMBEDDED CHIP
READY TO USE**

SMART CARD. SMART SOLUTION.

PIR SENSOR MODULE

PASSIVE INFRARED MOTION SENSOR



MOTION DETECTION

Detects human movement within range



HIGH SENSITIVITY

Reliable and accurate motion detection



ADJUSTABLE

Adjustable sensitivity and delay time



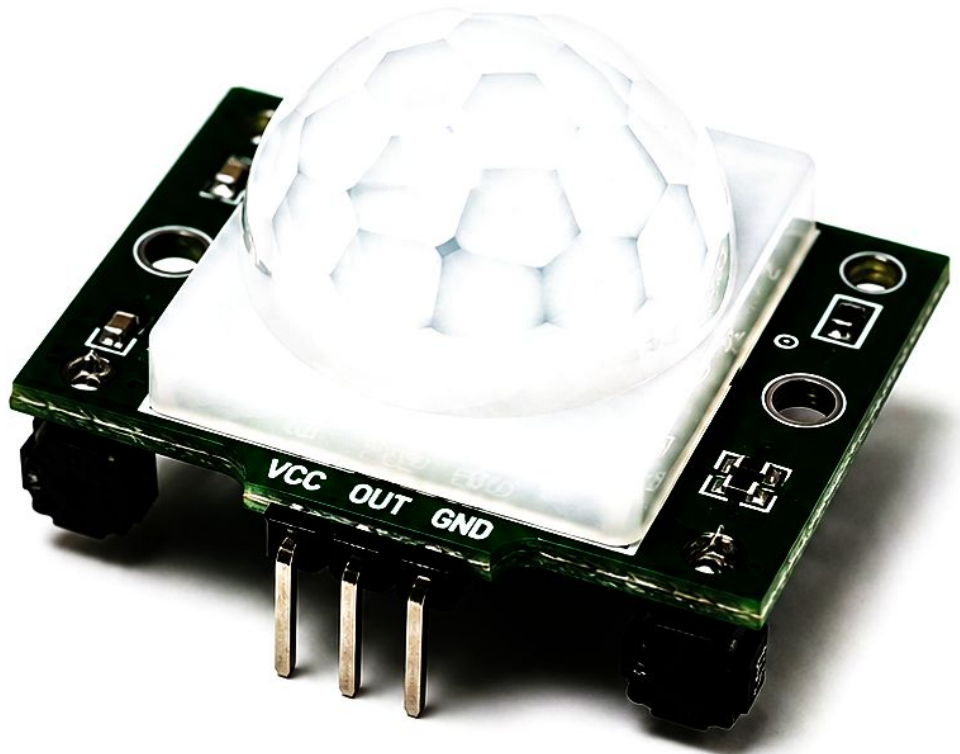
EASY TO USE

Simple 3-pin interface, easy to integrate



WIDE APPLICATIONS

Home automation, security systems, lighting, alarms



SPECIFICATIONS

- **Operating Voltage:** 4.5V – 20V DC
- **Quiescent Current:** < 65µA
- **Output:** High / Low (3.3V TTL)
- **Detection Range:** Up to 7 meters
- **Detection Angle:** ≤ 110°
- **Adjustable Delay Time:** 3 sec – 5 min
- **Adjustable Sensitivity:** Yes
- **Operating Temperature:** -20°C to +70°C
- **Dimension:** 32mm x 24mm x 18mm

PIN DESCRIPTION

PIN	NAME	DESCRIPTION
1	VCC	Power Supply (4.5V – 20V DC)
2	OUT	Digital Output (High/Low)
3	GND	Ground

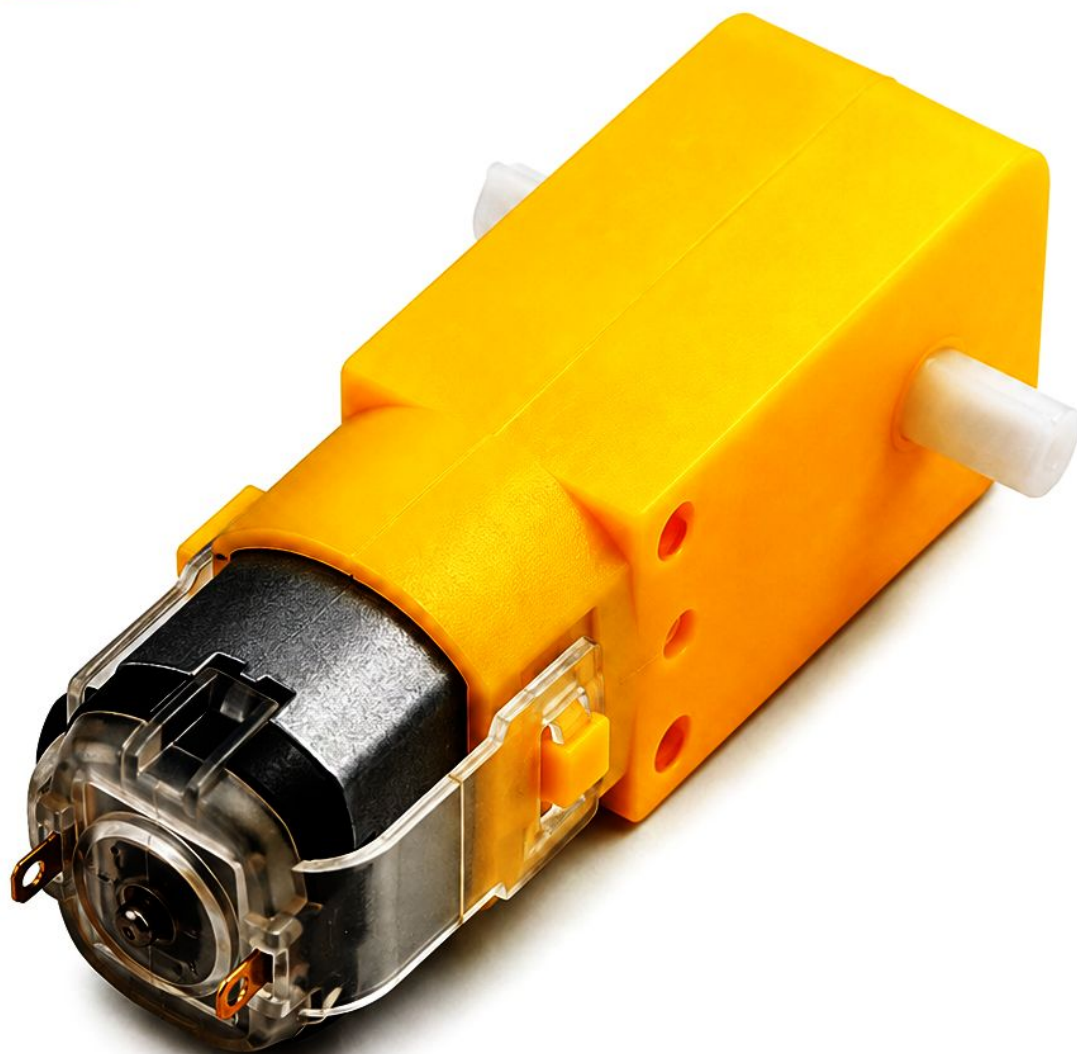


APPLICATIONS

- Motion Activated Lighting
- Security Alarm Systems
- Automatic Door Systems
- Home Automation
- Energy Saving Systems
- IoT Projects

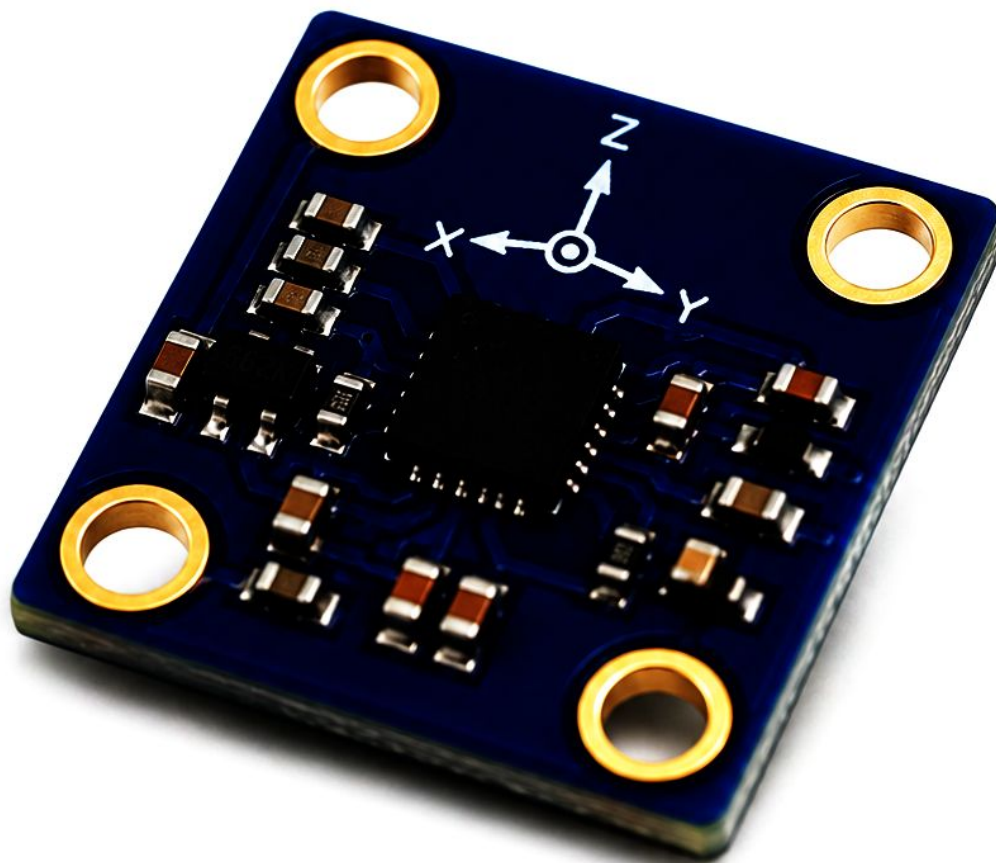


YELLOW BO MOTOR



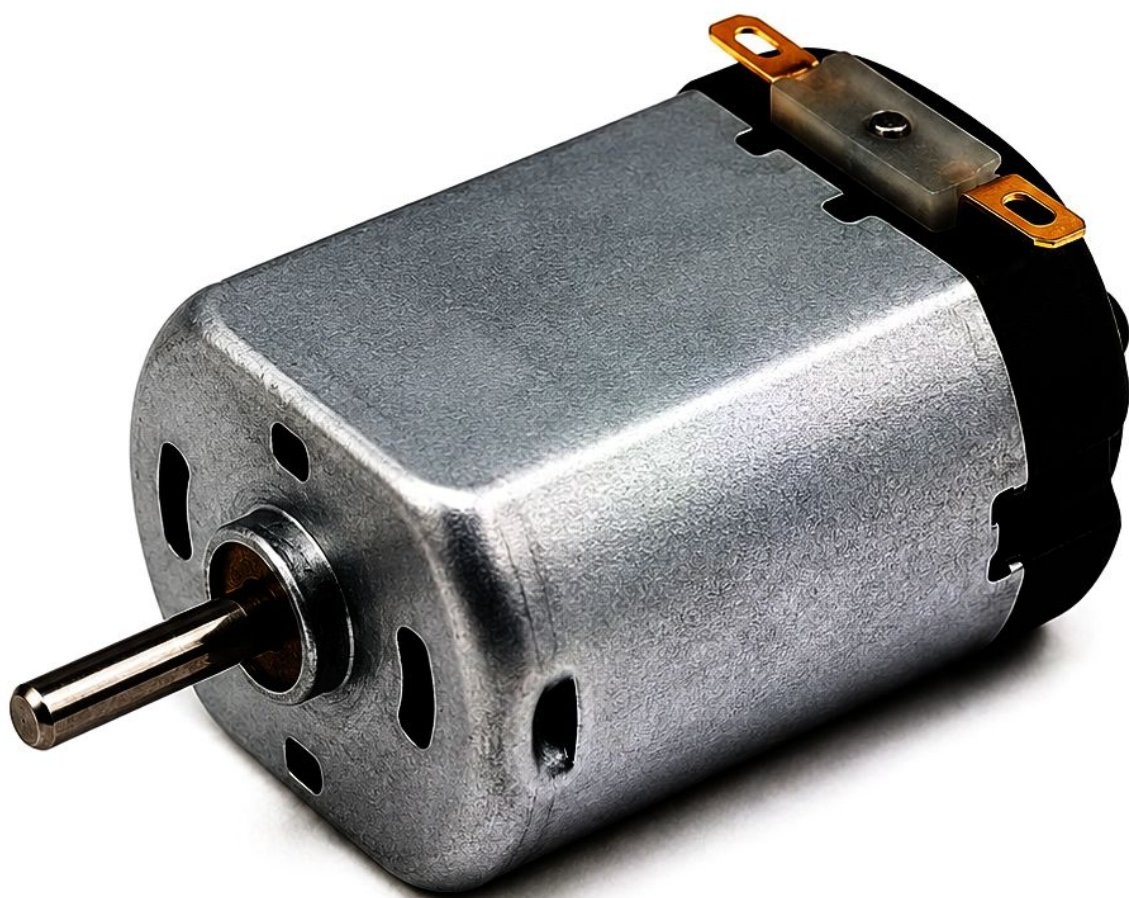


GYROSCOPE SENSOR





MINI MOTOR





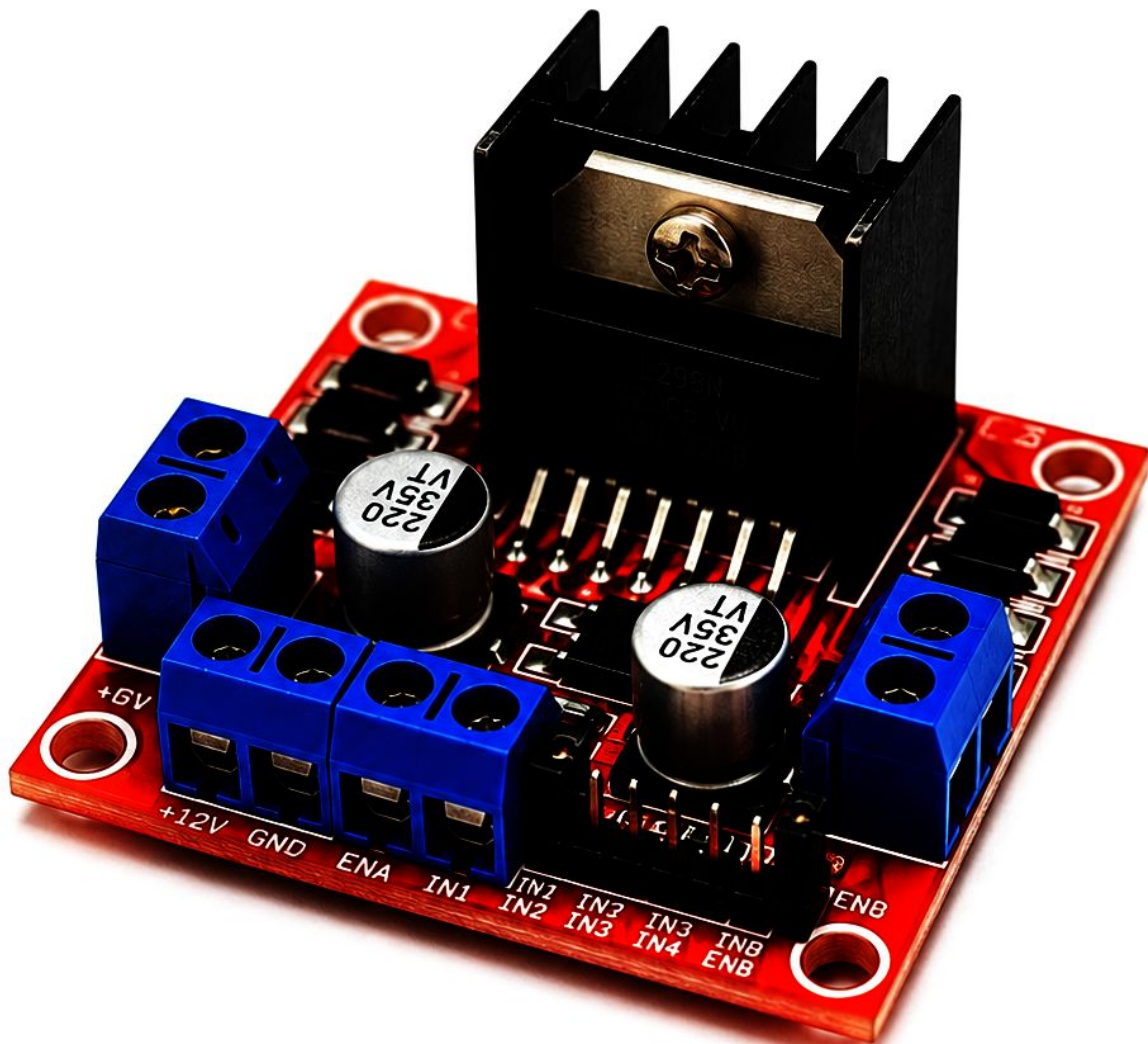
90 SERVO MOTOR





L298N

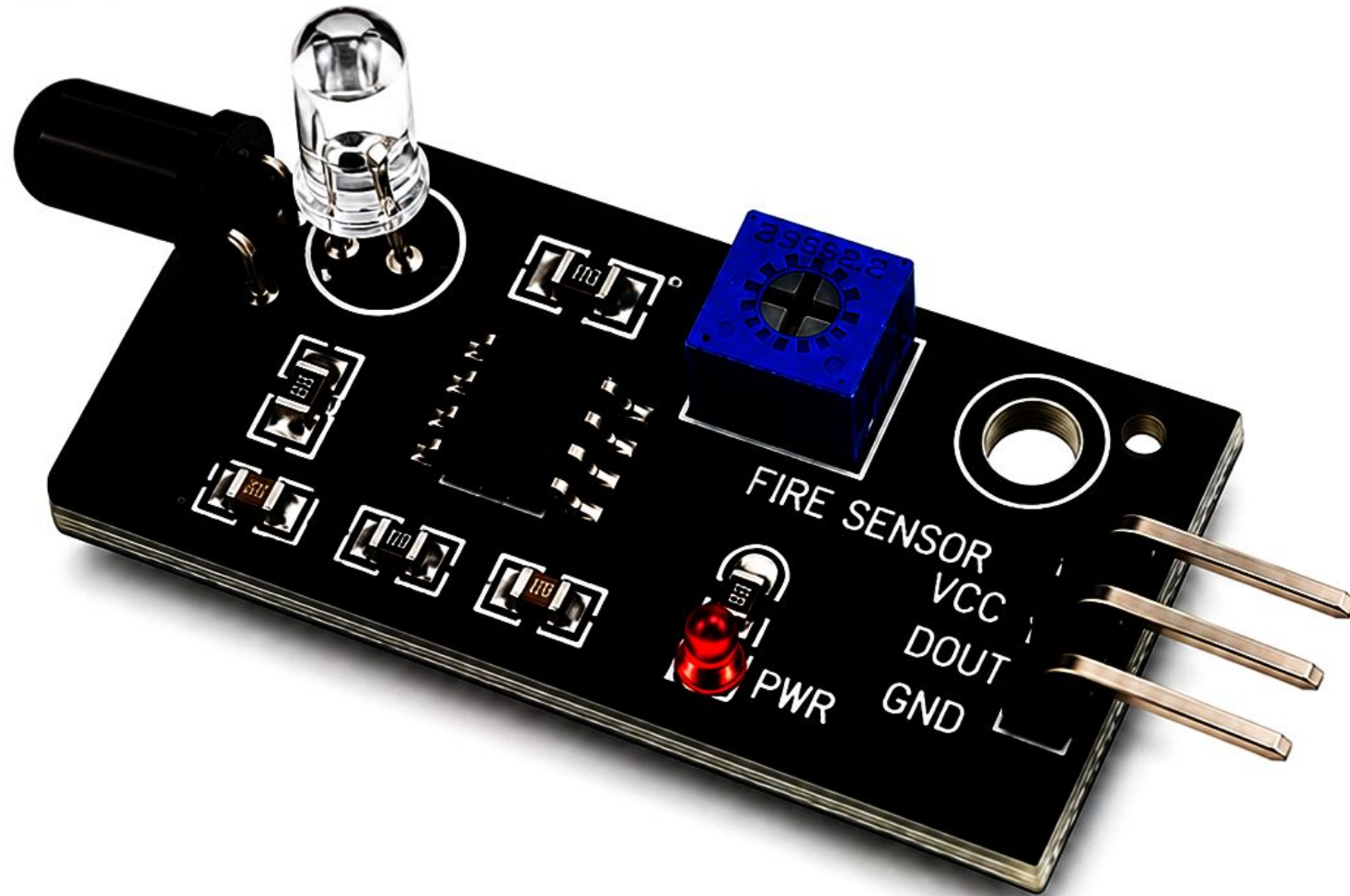
MOTOR DRIVER



MQ-6 GAS SENSOR MODULE



FIRE SENSOR MODULE



MQ-3 GAS SENSOR MODULE



MQ-2 GAS SENSOR MODULE





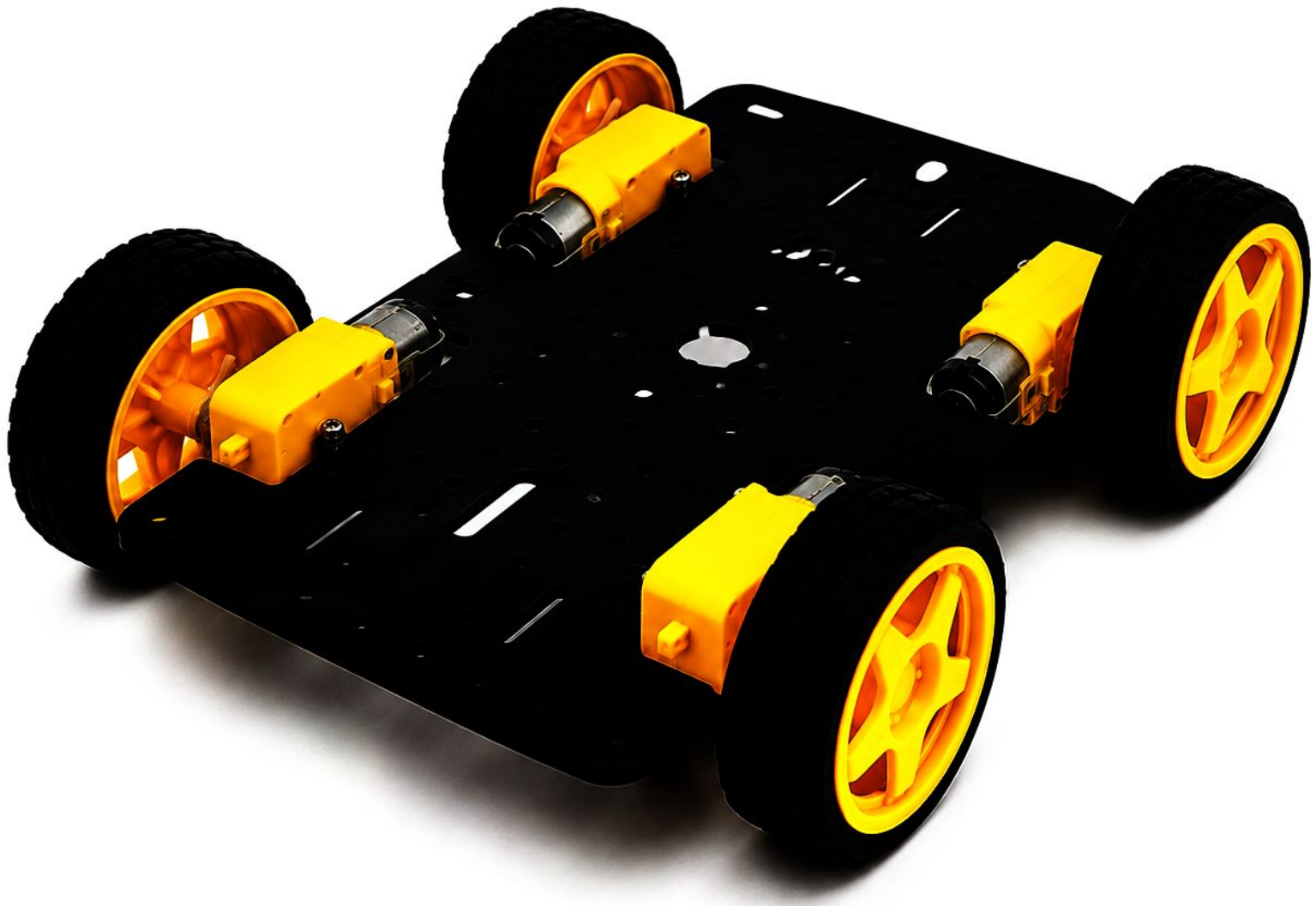
3.7V LITHIUM IRON PHOSPHATE BATTERY

-  Type
LiFePO4
-  Nominal Voltage
3.7V
-  Capacity
1500mAh
-  High Safety
Long Life
-  Eco Friendly
Rechargeable



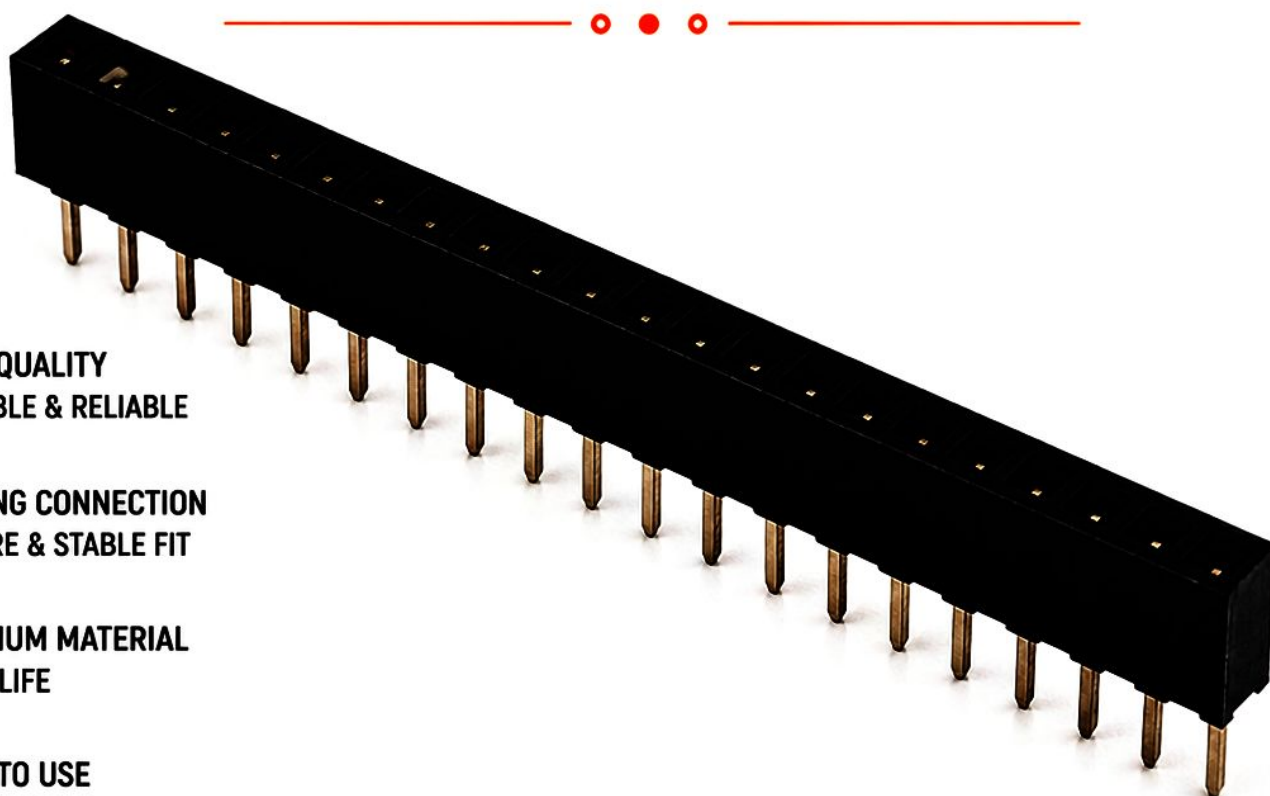


4 WHEEL CHASSIS





FEMALE PIN CONNECTOR



HIGH QUALITY
DURABLE & RELIABLE



STRONG CONNECTION
SECURE & STABLE FIT



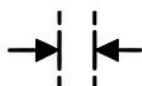
PREMIUM MATERIAL
LONG LIFE



EASY TO USE
PLUG & PLAY



TYPE
FEMALE



PITCH
2.54mm



ROWS
1 ROW



MATERIAL
HIGH QUALITY
PLASTIC

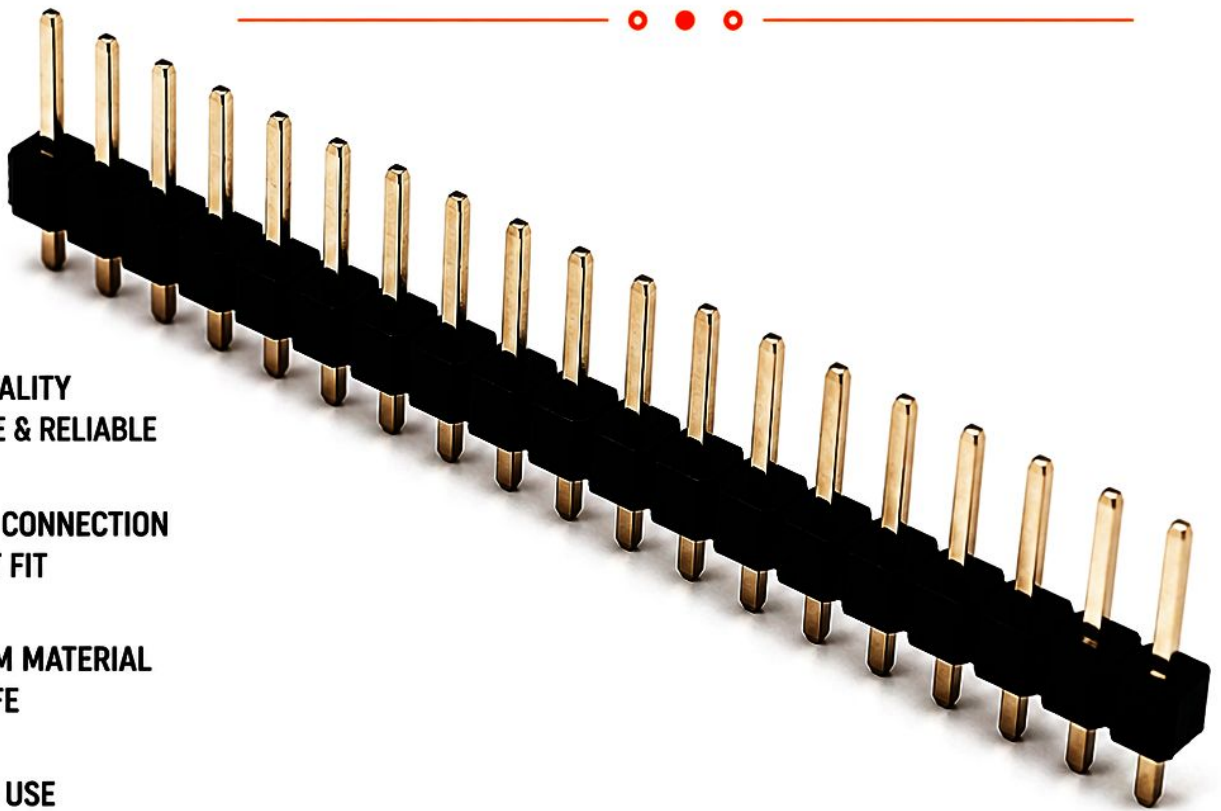
AVAILABLE OPTIONS

1x40 PIN | 1x20 PIN | 1x10 PIN

- VARIOUS LENGTH OPTIONS
- COMPATIBLE WITH 2.54mm MALE PINS



MALE PIN CONNECTOR



HIGH QUALITY
DURABLE & RELIABLE



STRONG CONNECTION
PERFECT FIT



PREMIUM MATERIAL
LONG LIFE



EASY TO USE
PLUG & PLAY

AVAILABLE OPTIONS

1x40 PIN | 1x20 PIN | 1x10 PIN



2.54mm

PITCH
2.54mm



TYPE
MALE



MATERIAL
BRASS PIN



RELIABLE
& STURDY



3.6V LITHIUM ION BATTERY

• • •



2 WHEEL CHASSIS

WITH 2 MOTORS & 2 CELL BATTERY HOLDER

